Detecting and quantifying wash trading of non-fungible tokens

Jonathan Chow; Singapore Management University; hartonot.2019@scis.smu.edu.sg

Abstract: Wash trading is a process where a trader buys and sells a security for the express purpose of feeding misleading information to the market; it is illegal in traditional markets and has since been made illegal for NFT trades. The literature on NFT wash trading detection primarily refers to identifying strongly connected components; however, this can be bypassed by taking advantage of the anonymous nature of a blockchain. A trend where the top NFT collections by market cap see a much-reduced wash trading percentage strongly indicates the presence of hidden wash trades. To combat this, this study uses the user transaction graph of the native Ethereum tokens alongside typical asset transfer graphs to create three different wash trading detection algorithms.

1.0 Introduction

Since the inception of Bitcoin in 2009, the world has been introduced to the concept of a decentralised and distributed ledger that allows multiple parties to have a secure and transparent record of transactions without the need for a central authority, blockchains.

In 2015, smart contracts were pioneered as part of the Ethereum blockchain. Smart contracts are computer programs stored on the blockchain that executes when triggered by a transaction from a user.

In 2018, non-fungible tokens (NFTs) were popularised in the Ethereum Improvement Proposal, ERC-721. (Entriken, 2018) In contrast to fungible tokens, where each token is identical and can be exchanged on a one-to-one basis, non-fungible tokens are unique and indivisible digital assets.

```
function balanceOf(address _owner) external view returns (uint256);
function ownerOf(uint256 _tokenId) external view returns (address);
function safeTransferFrom(address _from, address _to, uint256 _tokenId, bytes data) external payable;
function safeTransferFrom(address _from, address _to, uint256 _tokenId) external payable;
function transferFrom(address _from, address _to, uint256 _tokenId) external payable;
function approve(address _approved, uint256 _tokenId) external payable;
function setApprovalForAll(address _operator, bool _approved) external;
function getApproved(uint256 _tokenId) external view returns (address);
function isApprovedForAll(address _owner, address _operator) external view returns (bool);
```

Figure 1: ERC-721 interface (ethereum.org)

Everything on a blockchain is a transaction initiated by a user, and as the name suggests, groups of transactions form a block to create a consensus. The NFT is no different. The smart contract is created with codes added as a parameter to the deployment transaction. And given that a user wishes to send his NFT to a friend, he would create a transaction with his friend's wallet address. As such, seeing an NFT's asset transfer graph is as simple as viewing the NFT by its transaction history.

The asset transfer graph is a graph of a singular asset's transaction history, where an asset is defined by its unique composite identifier of smart contract address and token id, the nodes represent a user (or wallet), and the edges represent asset transfers between them. This is distinct from a user transaction graph, a graph of all transactions between users, where different types of transactions can exist on the same graph.

On decentralised NFT marketplaces, the seller lists the NFT on the platform, and the buyer gets to fulfil the listing. Buyers can also create buy orders or "offers" for the NFT, and the owner can choose to fulfil the request. Both buy orders and listings are converted into offers and consideration in the decentralised NFT marketplace introduced by Open Sea, Seaport, which is a source of data for this study.

1.1 Prevalence of NFT wash trading

Wash trading is a process where a trader buys and sells a security for the express purpose of feeding misleading information to the market. (Chen & Anderson, n.d.)

NFT wash trading volume peaked close to January 2022, where it was 11.56B, and as of February 2022, wash trading made up 23.4% of unadjusted trading volume across all marketplaces. Wash trading is more prevalent on NFT marketplaces that provide token incentives for NFT trades, where wash trading amounted to up to 85% of unadjusted trading volume on X2Y2 in the same month. (Total NFT Wash Trading, 2022-2023, n.d.)

The collection with the most wash trading volume is Terraforms by Mathcastle, with 12.1B, 4.7B of which are made by the top 2 wallets by wash trades. (footprint analytics, 2023) This is only possible because there are no royalties on this collection, and the only cost of wash trading is the nominal gas fees incurred by the transaction. Many of the top collections by wash trading volume also have no royalties on the collection.

On top of token incentives and royalty-free collections, the rampant wash trading can be attributed to metrics such as trading volume and floor price being typical indicators for a collection's ranking in an NFT marketplace, where the value of NFTs is generally judged by the price and volume of trades of both the asset in question and that of its collection.

2.0 Related Work

In this section, an examination is done of the existing literature on NFT wash trading detection, and there are two schools of thought. The most prevalent detection method is identifying strongly connected components in the NFT's asset transfer graph. In contrast, open-source research uses more rudimentary yet effective methods we can learn from. Lastly, looking outside of NFT trading, the literature on wash trading on decentralised exchanges and traditional finance can inform the efficacy of the techniques used in NFT wash trading.

2.1 SCC Detection

A directed graph is strongly connected if there is a directed path from each vertex to every other vertex in the graph. Detecting strongly connected components (SCC) of a graph (Jain, 2023), primarily when multiple directed cycles exist within an SCC, strongly indicates foul play and can be brought on by a wash trader using multiple accounts in purchasing their own NFT listing.

There are few studies that have focused on NFT wash trading detection. Two separate implementations approached NFT wash trading detection with simple SCC detection, and one implementation attempted to use a form of intention discovery to aid in the detection.

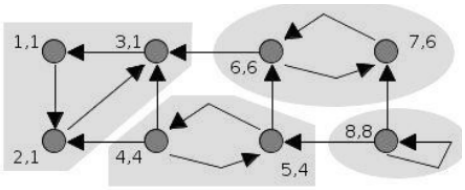


Figure 2: Graph with strongly connected components marked (wikipedia)

Typically, SCC-based NFT wash trading detection searches through a graph of the transfer history of assets in an NFT collection for a set of SCC (or sets or vertices). Tarjan's algorithm (Jain, 2023) is one algorithm used to find a set of SCCs by scanning the directed graph of the asset's trading history or asset transfer graph.

In a particular study, an SCC is flagged as a set of wash trades when more than five cycles are found in it. This study analysed 18 NFT collections with small trading volumes and had a true positive rate of 87% and a true negative rate of 100% when compared to a third-party analysis. Issues this paper faces include limited data access and, long loading times, and the readability of graphical visualisations. (Jung, 2022, #)

A second study also used a similar SCC detection method but did it on top of both an asset sales graph and an asset transfer graph to more reliably detect cycles from the graph. In addition, the paper tracks changes in prices of NFT listings and the profits made from a trade or sets of trades, which was used as a helpful indicator for a wash trade. Similar to an asset transfer graph, an asset sales graph is a graph of all NFT sales between wallets, and was used to fill in the missing edges on the asset transfer graph. (Liu, 2023, #)

2.2 Open-source Research

Open-source research from platforms like Dune, Footprint Analytics, and Chainalysis are widely cited due to their transparency, ease of use, and editable queries for data aggregates. (footprint analytics, 2023)

Dune's "Ethereum NFTs Wash Trading" research (hildobby, 2022) explained the mechanics behind its wash trading detection in an article and broke it down into five filters. The first three filters are a subset of SCC detection that detected cycles in the data, whereas the fourth and fifth used data that was unavailable in the studies mentioned above. The five filters are as follows:

1. Buyer equals seller, creating a self-loop.
2. Back-and-forth trades, creating a loop between 2 users or vertices.
3. Purchasing an NFT 3 times, which would necessarily mean creating a cycle unto itself due to the non-fungibility of ERC721 NFTs.
4. Buyer and seller being first funded with the same wallet, which creates a non-directed cycle between 2 accounts.
5. Flash loan transactions used, which indicates the intention of not spending money in the trade.

2.3 DEX and TradFi Wash Trading

Looking at research on wash trading in decentralised exchanges (DEXs) and traditional finance (TradFi), there is evidence of SCC detection in multiple papers to find potentially colluding traders from the user transaction graph.

There are also many studies that used trade submission interval matching, where the length of time between the matched buy and sell orders is used as a key metric in determining wash trades. Typically, wash trades have tighter submission intervals on matching orders to minimise the risk of another individual unintentionally picking up the order. (Victor, 2021, #)

And lastly, volume matching is also commonly used to detect wash trades, by matching up sets of buy orders to sets of sell orders of precisely the same volumes. (Cao, 2016, #) Unfortunately, volume matching as a methodology does not work in NFT wash trading detection due to the non-fungibility of NFTs.

3.0 Problem Formulation

Wash trading detection on NFT trades and transfers primarily uses a detection algorithm that identifies SCCs. However, this can be bypassed by users easily due to the anonymous nature of a blockchain, where users typically create multiple accounts to perform wash trading. The SCC detection method requires a user to reuse an account one or more times to detect the wash trading of that asset.

There is a trend where the top NFT collections by market cap see a much-reduced wash trading percentage compared to both the average wash trading percentage and the wash trading percentage of the top NFT collections by trading volume. (footprint analytics, 2023) This is indicative of some hidden wash trades on more valuable collections and that the wash trading we do detect is from users who are not putting in the effort to mask their trades.

In this study, to better detect masked wash trading of NFTs, the user transaction graph of ETH (native Ethereum tokens) is used to find transactions that connect otherwise separate wallets used for wash trading an NFT. Despite references to using the user transaction graph and analysing fungible token transactions in existing research, to the best of our knowledge, there was no study done to take advantage of learning user behaviour from the user transaction graph to better inform the detection of NFT wash trades.

4.0 Methodology

In this section, the overarching goal of optimising for the detection of NFT wash trades is reflected from the data selection process, to the wash trading detection. The methodology expands from the current use of SCCs to find more structures in the graphs that is indicative of wash trades.

4.1 Data Selection

Time Period. In this study, the data is scrapped from either an API provider or a JSON RPC. Seeing the origin block of Ethereum dated on the 30th of July 2015, a plot of NFT trading volume, no. of NFT transactions, and unique trading wallet addresses (UAW) over time helps select an optimal time period over which to scrape data.

The time period selected, highlighted in blue in Figure 3, is between 1 May 2021 to 1 June 2023. Looking at the data, the time period averaged 40.6M in daily trading volume and 60,351 in daily transactions. This compares to an average of 0.5M and 1,332 in daily trading volume and transactions respectively for the two years prior to the time period.

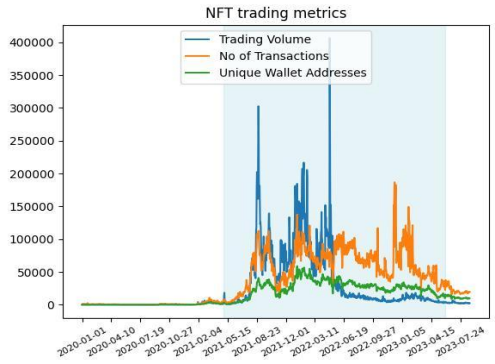


Figure 3: NFT trading metrics between 2020 and 2023 (nftscan)

NFT Collections. The top 20 ERC-721 NFT collections, rank ordered by trading volume was selected for this study; because collections with higher trading volumes would better accentuate the effects of wash trading. The following are the selected collections:

Terraforms, Meebits, BoredApeYachtClub, Dreadfulz, CryptoPunks, MutantApeYachtClub, dotdotdot, More Loot, Azuki, Otherdeed, Hashes, CloneX, Art Blocks, Moonbirds, MineablePunks, Wrapped Cryptopunks, Doodles, Loot, BoredApeKennelClub, Beanz

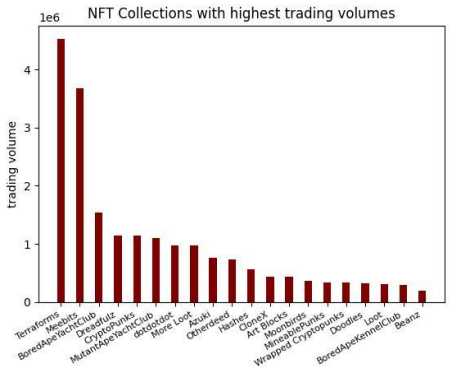


Figure 4: Top 20 NFT collections by trading volume (nftscan)

Datasets. The Moralis API provider (moralis, n.d.) is used to scrape data for datasets 1, 2, and 3, which are actively used in the detection algorithms in this paper. The Footprint Analytics API provider (footprint analytics, 2023) which was used for datasets 4, 5, and 6, are used for the evaluation of the detection algorithms. The datasets amounts up to 12.7M data points for the two-year and one-month period of 1 May 2021 to 1 June 2023.

The ETH transfers dataset has been reduced to the transfers of all wallet addresses that participated in either an NFT trade or transfer of the Terraform NFT collection during the aforementioned time period. This reduced the dataset from an approximated 323M data points to 9.3M data points.

No.	Dataset	No. of Rows	Details
1	NFT Trades	547,361	Transaction Details
2	NFT Transfers	2,686,773	Transaction Details
3	ETH Transfers	9,299,772	Transaction Details
4	FA Suspicious Addresses	104,857	address, trading_value, trades
5	FA Tainted Tx	104,857	tx_hash
6	FA WashTrade Ratios	100	total_volume, wash_trading_volume, wash_trading_percentage

Table 1: List of Datasets used

The following datasets are filtered for transactions during the aforementioned time period:

1. **Dataset 1** is a dataset of all NFT Trades on the Open Sea NFT Marketplace for the top 20 NFT collections by trading volume
2. **Dataset 2** is a dataset of all NFT Transfers for the top 20 NFT collections by trading volume
3. **Dataset 3** is a dataset of all ETH Transfers to and from all wallet addresses that participated in either an NFT trade or transfer of the Terraform NFT collection (the number one collection by trading volume during the time period)
4. **Dataset 4** is a list of wallet addresses suspected of wash trading, created by Footprint Analytics
5. **Dataset 5** is a list of wash trades, created by Footprint Analytics
6. **Dataset 6** is a list of the top 100 collections by transaction volume across all time, with its trading volume, and wash trade percentage during the time period, created by Footprint Analytics

Collection	NFT Trades	NFT Transfers	Collection	NFT Trades	NFT Transfers
Terraform	13,152	55,141	Hashes	1,071	5,855
Meebits	21,466	76,900	CloneX	23,006	134,551
BoredApeYachtClub	19,777	199,283	Art Blocks	199,165	518,174
Dreadfulz	13,341	43,506	Moonbirds	11,476	71,690
CryptoPunks	0	27,242	MineablePunks	3,505	10,530
MutantApeYachtClub	39,321	271,260	Wrapped Cryptopunks	18	20,153
dotdotdot	10,380	20,495	Doodles	26,255	118,746
More Loot	6,190	153,297	Loot	12,442	34,361
Azuki	28,898	131,332	BoredApeKennelClub	25,515	113,714
Otherdeed	64,273	506,076	Beanz	28,110	174,467

Table 2: Breakdown of datasets 1 and 2 by collection

4.2 Data Preprocessing

NFT transfers. The data from NFT transfers consists of both a from and to address, as well as the token identifier, which is the smart contract address and token id combined. The details of the NFT transfer is used to build a universal interface for an asset transaction, "Transaction Details". The universal interface is used by datasets 1, 2, and 3, which are backfilled by dummy data for fields that do not apply to the dataset.

Notably, (1) the field "token_ids" is filtered and processed for multiple ids during the preprocessing stage, (2) seller and buyer addresses are also used as from and to addresses respectively, (3) the price token address is left as an empty string for the native Ethereum token (ETH), (4) the price value doesn't record the decimals as part of the dataset, and (5) the block number is converted from a hexadecimal to a decimal value during preprocessing.

```
TransactionDetails = {
  address: '0x4e1f41613c9084fdb9e34e11fae9412427480e56',
  token_ids: 6493,
  seller_address: '0x1bcb63e4dc7ea87561faf8dfa271f4a6ee60bcc1',
  buyer_address: '0x69efd597a3f7018ce35f38f53ba05d42c4f6b1c3',
  marketplace_address: '0x00000000000000adc04c56bf30ac9d3c0aaf14dc',
  price_token_address: '',
  price: 1990000000000000000,
  block_number: 17321906,
  tx_hash: '0x61ef0a728ab2091d99554f9cfe0c39d4dd403616df0932fcb5914025bb7d4857'
}
```

Figure 5: An example of Transaction Details

NFT trades. The raw data for NFT trades comes from the Seaport contract event, Order Fulfilled, which comes in the form of an array of assets offered, and an array assets for payment or "consideration", this is used for both listings and buy orders alike. Using the Moralis API provider, the data is preprocessed to have an array of assets offered, all from the same collection, and a singular payment token for consideration.

The NFT trades dataset, is then processed like the NFT transfers dataset, where each datapoint creates a Transaction Details object that is fed to the graph. Unfortunately, due to a lack of data, the datasets 1 and 2 are unable to be combined seamlessly and would be used as 2 separate datasets during this report.

ETH transfers. After filtering out for zero value ETH transfers, the dataset is then processed like the NFT transfers dataset, where each datapoint creates a Transaction Details object that is fed to the graph.

4.3 SCC Detection

The data model created is as depicted in Figure 6. After preprocessing the data to create a list of Transaction Details, it is fed into the data model / graph. Each Transaction Detail fed into the data model creates an instance of the Transaction object, which creates an instance of the sender, receiver and asset of the transaction. The Asset object stores its transaction history as an array of Transaction objects, while the User object stores a list of outgoing and incoming transactions.

To create the asset transfer graph from the Asset's transaction history, (1) sort the transaction history by its block number in ascending order, then (2) for each transaction create 2 nodes, a sender and receiver and connect them together. If the previous transaction's receiver is not the current transaction's sender, connect a directed edge from the receiver to the sender. This will solve issues of missing data in the graph.

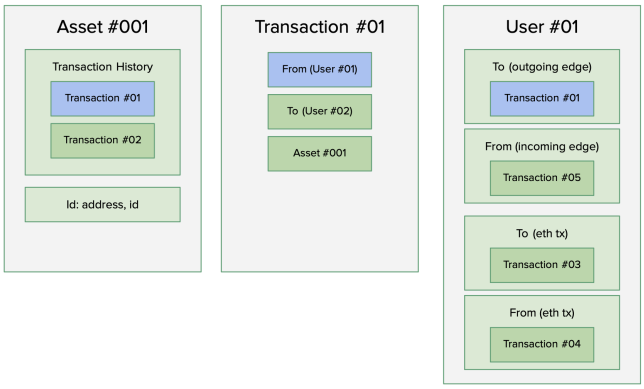


Figure 6: Data model with an Asset Transfer Graph and User Transaction Graph

The data model has both an asset transfer graph and a user transaction graph that can be used together or separately as needed. Having created the graph, we can start running our wash trading detection algorithm.

```
algorithm tarjan is
    input: graph G = (V, E)
    output: set of strongly connected components (sets of vertices)

    index := 0
    S := empty stack
    for each v in V do
        if v.index is undefined then
            strongconnect(v)

    function strongconnect(v)
        // Set the depth index for v to the smallest unused index
        v.index := index
        v.lowlink := index
        index := index + 1
        S.push(v)
        v.onStack := true

        // Consider successors of v
        for each (v, w) in E do
            if w.index is undefined then
                // Successor w has not yet been visited; recurse on it
                strongconnect(w)
                v.lowlink := min(v.lowlink, w.lowlink)
            else if w.onStack then
                // Successor w is in stack S and hence in the current SCC
                // If w is not on stack, then (v, w) is an edge pointing to an SCC already found and must be ignored
                // The next line may look odd - but is correct.
                // It says w.index not w.lowlink; that is deliberate and from the original paper
                v.lowlink := min(v.lowlink, w.index)

        // If v is a root node, pop the stack and generate an SCC
        if v.lowlink = v.index then
            start a new strongly connected component
            repeat
                w := S.pop()
                w.onStack := false
                add w to current strongly connected component
            while w ≠ v
            output the current strongly connected component
```

Figure 7: Tarjan's algorithm

Figure 7 helps give a better understanding of Tarjan's algorithm, which starts at a random vertex and runs a depth-first search on it. It saves a new low-link value for all vertices in a "link" when detecting a "back-edge". The cycle then repeats for another random unvisted vertex until all vertices are visited. The algorithm returns with a set of SCCs.

The SCC detection algorithm is run iteratively for each asset as defined by the composite identifier of smart contract address and token id on the relevant asset transfer graph.

To prevent an abundance of false positives, our detection algorithm requires a SCC to have at least 2 cycles to be considered a wash trade. To count the number of simple cycles, simply count the number of "back-edges" by comparing the low-link value to the index value.

The key metric is the number of wash trades. Hence, for all SCCs that are flagged, or have at least 2 cycles, every edge in the SCC counts as a single wash trade. This can be divided against all edges in the graph to get the wash trade share, which is the percentage of trades detected as wash trades.

4.4 Wash Trading Detection

Beyond the SCC detection on a simple asset transfer graph, in this paper we attempt 3 types of wash trading detection algorithms as follows: (the acronyms SCC A, SCC AU, and SCC AC will be used to more clearly denote the detection methods in this paper)

1. SCC detection on an asset transfer graph (SCC A)
2. SCC detection on an asset transfer graph and user transaction graph (SCC AU)
3. SCC detection on an asset transfer graph and simple cycle detection on an undirected user transaction graph (SCC AC)

SCC detection on an asset transfer graph (SCC A). To get the wash trading aggregates for each NFT collection, we can run the algorithm on the data of each NFT collection independently.

SCC detection on an asset transfer graph and user transaction graph (SCC AU). The graph of the combined asset transfer graph and user transaction graph includes a single asset's transaction history, and all ETH transactions. An iteration of the algorithm on all assets in the Terraform NFT collection is done and the results of the number of wash trades and total trades are aggregated at the end.

A slight adjustment in the depth-first search used in Tarjan's algorithm includes the termination of all branches that only contains ETH transfers at a depth of 30; and the pruning of all SCCs where an asset transfer edge does not exist from the result. Naturally, the algorithm only counts asset transfer edges and not ETH transfer edges.

SCC detection on an asset transfer graph and simple cycle detection on an undirected user transaction graph (SCC AC). The combined asset transfer graph and user transaction graph is used, but independently of each other. First, the standard SCC detection is done on the asset transfer graph.

Then, for each node of each asset transfer graph, a breath-first search is done on an undirected user transaction graph of ETH transfers to a depth of 2. Where a ETH transfer to or from a wallet / user in the asset transaction graph connects to another wallet / user in the same asset transaction graph, all asset transfers between the 2 wallets / users are considered wash trades. The union of the wash trades detected from both the SCC detection method and the simple cycle detection method is the result of the method SCC AC.

The choice of a doing a breath-first search to a depth of 2 was because the user transaction graph in this study is limited to the transfers of all wallet addresses that participated in either a NFT trade or transfer of the Terraform NFT collection during the aforementioned time period, and so the breath-first search to a depth of 2 would find all 2-deep cycles present in a full user transaction graph if the transaction is not terminated by the time period. Figure 8 more clearly describes the 2-deep cycles mentioned here.

5.0 Evaluation

In this section, the evaluation datasets 4, 5, and 6 are used to evaluate the three detection methods created, SCC A, SCC AU, and SCC AC respectively. This companied with a discussion on the theory behind the three detection methods helps illustrate the strengths and weaknesses of each method.

5.1 Results

The three detection methods are aggregated with key data metrics from each individual SCC or simple cycle, namely (1) number of SCCs, (2) number of nodes, (3) number of edges, and (4) number of flagged edges. The total number of flagged edges against the total number of asset transfer edges gives the ratio coined the Wash Trade Share; which is presented in Table 3 for detection method SCC A. Note that Wash Trade Shares are displayed as percentages below and that flagged edges are only flagged if they are asset transfer (or trade) edges.

Collection	Wash Trade Share (Dataset 1)	Wash Trade Share (Dataset 2)	FA's Wash Trade Share
Terraform	8.37	45.89	99.76
Meebits	7.82	34.51	96.02
BoredApeYachtClub	1.01	9.78	1.46
Dreadfulz	8.30	78.97	98.45
CryptoPunks	0.00	0.05	0.61
MutantApeYachtClub	2.75	6.17	1.46
dotdotdot	7.02	35.26	99.12
More Loot	0.37	28.27	99.50
Azuki	5.22	21.05	0.54
Otherdeed	2.80	3.41	3.95
Hashes	9.17	58.27	NA
CloneX	1.17	0.27	0.81
Art Blocks	15.37	59.03	15.58
Moonbirds	9.90	53.41	NA
MineablePunks	10.70	58.27	59.56
Wrapped Cryptopunks	6.21	9.92	NA
Doodles	8.61	10.4	0.82
Loot	15.55	96.03	73.43
BoredApeKennelClub	5.58	7.73	4.38
Beanz	13.05	17.33	2.52

Table 3: Wash Trade Share of all collections using method SCC A

The detection methods SCC AU and SCC AC were only run against the Terraform(s) NFT collection due to issues with data collection of the user transaction graph.

Method	Wash Trade Share (Dataset 1)	Wash Trade Share (Dataset 2)	FA's Wash Trade Share
SCC A	8.37	45.89	99.76
SCC AU	16.19	51.25	
SCC AC	9.68	64.41	

Table 4: Wash Trade Share of Terraforms for all detection methods

Missing Data was also tabulated and amounted to up to 3% in the NFT trades dataset (dataset 1), and amounted up to 0.5% in the NFT transfers dataset (dataset 2). Dataset 3 was not evaluated for missing data. This percentage is calculated from the asset transfer graph, where any two adjacent transactions in a time-sorted asset's transaction history has to have the first transaction's receiver have the same address as the second transaction's sender.. This is notably only a subset of missing data.

5.2 Third Party Analysis

Aside from Footprint Analytics' wash trade share data, datasets 4 and 5 provides the percentage of suspicious wallet addresses flagged and percentage of suspicious transaction data flagged.

Filtering out addresses and transactions not found on either the data model or datasets 4 and 5 respectively. The percentage of total number of flagged data detected by the three detection methods against Footprint Analytics' flagged data list are as follows. Only dataset 2 was used in this analysis.

Method	Suspicious Wallet Addresses Flagged (%)	Suspicious Tx Flagged (%)
SCC A	77.98	85.10
SCC AU	95.03	89.23
SCC AC	71.33	84.40

Table 5: Comparison against Footprint Analytics' data on suspicious addresses and transactions

5.3 Discussion

In this discussion, a baseline is first created by discussing Table 3's results from the detection method SCC A. Then a comparison between detection methods SCC AU and SCC AC is made, before diving into the theory behind method SCC AC. Finally, we will look to the 3rd party analysis to wrap up our discussion.

Baseline with method SCC A. In Table 3, a baseline for how our datasets work is provided against a reputable research's findings, using only detection method SCC A.

Dataset 1 has a much lower detection percentage across the board for all collections, and can be attributed to it being a subset of dataset 2, and / or to bad data collection. Dataset 1 was initially intended to meld with dataset 2, but due to a faulty dataset, an attempt to meld the data increases the missing data percentage to 65% from 0.5%. A calculated guess is that a address translation error due to proxies or internal transactions might have been the cause of the confusion.

Dataset 2's cosine similarity against Footprint Analytics' wash trade share is 0.63, which is significantly better than that of database 1's cosine similarity of 0.82. The cosine similarity value might still be high, but it would be notably skewed in such a small dataset from random fluctuations in data. The cosine similarity value shows that there is still a decent correlation, despite the large differences in results.

Dataset 2's general trend of a lower wash trade share compared to Footprint Analytics' data aggregates, can be attributed to either differences in the dataset or the detection method. Footprint Analytics has not opted to reveal its detection methods and hence it is possible that the research did not use the typical SCC method, and if they did, perhaps it only required one cycle to flag a SCC.

Comparison of SCC AU and SCC AC. In Table 4, the detection methods ordered by the number of wash trades detected from lowest to highest are SCC A, SCC AU, and SCC AC, for dataset 2. It is trivial that SCC AU and SCC AC have higher detection values than SCC A as it is a subset of the other 2 detection methods. However, it is interesting that method SCC AC boasts a higher detection rate compared to method SCC AU, considering that SCC AU runs a depth-first search of depth 30 compared to SCC AC which runs a breath-first search depth of 2 (on the user transaction graph).

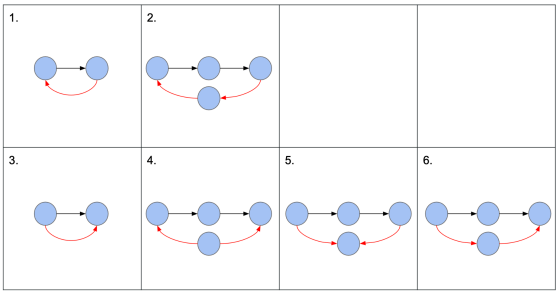


Figure 8: Simple cycles with ETH transfers of max depth 2

Theory of method SCC AC. Figure 8 displays all simple cycles created by ETH transfers and at least one asset transfer of up to a depth of 2, given that all edges are undirected and that a cycle is not created in the asset transfer graph. The nodes represent a wallet or a user, black arrows represent an asset transfer transaction, and a red arrow represents a ETH transaction.

Method SCC AU detects the simple cycle 1 and 2 in Figure 8. Whereas method SCC AC detects simple cycles 1 through 6. Simple cycle 4, which was used in Dune's research indicates a common source of money for the 2 accounts, and is a very common practice amongst wash trades, but is not detectable by conventional SCC techniques. Simple cycles 3, 5, and 6 show a user possibly infusing money from an account that made profit on a wash trade to another.

The higher detection rate of method SCC AC compared to method SCC AU indicates the relatively higher percentage of wash trades with short simple cycles on undirected edges, compared to longer directed cycles created by ETH transactions.

3rd party analysis. Last but not least, there was a decent result on the third party flagged data analysis despite the bigger variation found on the wash trading ratios. The flagged transactions and addresses have an average of 86.24% and 81.44% detection respectively compared to the 46% of wash trade ratios.

This is believed to be from the aggressive removal of transactions and addresses that do not exist in both our graph and Footprint Analytics' dataset. This however does indicate that despite the efforts spent on preprocessing and sourcing of data, that more could be done. Creating my own Ethereum node for data scraping would certainly be the way forward in the future.

References

Cao, Y. (2016, 11). Detecting Wash Trade in Financial Market Using Digraphs and Dynamic Programming. *IEEE Transactions on Neural Networks and Learning Systems*, *27*(11). https://ieeexplore.ieee.org/document/7298451

Chen, J., & Anderson, S. (n.d.). *Wash Trading: What It Is and How It Works, With Examples*. Investopedia. Retrieved November 19, 2023, from https://www.investopedia.com/terms/w/washtrading.asp

Entriken, W. (2018, January 24). *ERC-721: Non-Fungible Token Standard*. Ethereum Improvement Proposals. Retrieved November 19, 2023, from https://eips.ethereum.org/EIPS/eip-721

footprint analytics. (2023, June 16). *V*. YouTube. Retrieved November 19, 2023, from https://www.footprint.network/research/nft/nft-wash-trade-stats/top-most-wash-trading-collections?series_date=past90days

hildobby. (2022, 12). *Ethereum NFTs Wash Trading (OpenSea, Blur, LooksRare, X2Y2...)*. Dune. Retrieved November 19, 2023, from https://dune.com/hildobby/nfts-wash-trading

Jain, S. (2023, June 6). *Tarjan's Algorithm to find Strongly Connected Components*. GeeksforGeeks. Retrieved November 19, 2023, from https://www.geeksforgeeks.org/tarjan-algorithm-find-strongly-connected-components/

Jain, S. (2023, October 30). *Strongly Connected Components*. GeeksforGeeks. Retrieved November 19, 2023, from https://www.geeksforgeeks.org/strongly-connected-components/

Jung, S. O. (2022, 12 06). Towards a Web Application to Detect and Analyze Wash Trading in NFT Collections on the Solana Blockchain. https://wwwmatthes.in.tum.de/file/19apj4nvq7a8m/Sebis-Public-Website/Student-Theses-Guided-Research/Current-Theses-Guided-Researches/TUMKolleg-Sebastian-O.-Jung/Sebastian%20Jung%20Thesis.pdf

Liu, D. (2023, 02 07). NFT Wash Trading Detection. arXiv:2305.01543v1

moralis. (n.d.). *Web3 Data API | Moralis Web3 Documentation*. Moralis docs. Retrieved November 19, 2023, from https://docs.moralis.io/web3-data-api/evm

Total NFT Wash Trading, 2022-2023. (n.d.). CoinGecko. Retrieved November 19, 2023, from https://www.coingecko.com/research/publications/nft-wash-trading-marketplaces?utm_campaign=PR%2BData%2BCampaigns&utm_medium=social

Victor, F. (2021, 02 13). Detecting and Quantifying Wash Trading on Decentralized Cryptocurrency Exchanges. arXiv:2102.07001v1